PTF ALL

EQUIP COOLING EXHAUST OFF Light Is On Switch At NORM - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Equipment Cooling Normal Exhaust Fan [M98]
Replace EQUIP COOLING EXHAUST Switch (Flight Deck) [S59]
Repair Wiring for D160
Replace EQPT COOLING EXHAUST FAN CONT NORMAL Circuit Breaker [C1435]
Repair Wiring D1624-Circuit Breaker C1435
Replace Normal Exhaust Fan Control Relay [R29]
Replace EQPT CLG EXH FAN PWR-NORM Circuit Breaker [C836]
Repair Wiring Between EQPT CLG EXH FAN PWR-NORM Circuit Breaker and Equipment Cooling Normal Exhaust Fan
Cycle EQPT COOLING LOW FLOW DETECT EXHAUST Circuit Breaker
Clean Equipment Cooling Airflow Sensor [M100]
Replace Equipment Cooling Airflow Sensor [M100]
Repair Wiring D164-Circuit Breaker C1434
Repair Wiring D164-D11418
Repair Wiring D164-L47 Terminals
Replace Equipment Cooling Smoke Auto Reconfiguration Timer Relay [R1185]

A. Description

(1) The EQUIP COOLING EXHAUST OFF light on the equipment cooling panel is on with the EQUIP COOLING EXHAUST switch on NORMAL.

(2) The EQUIP COOLING EXHAUST OFF light comes on when the normal exhaust fan does not pull enough cooling air through the equipment cooling ducts. The monitoring circuit in the exhaust low flow detector then detects an inadequate air flow condition and provides a ground to allow the light to come on.

(3) This fault can also occur when power is first applied to the aircraft, the equipment cooling system is started, and these conditions occurred:

(a) The aircraft was parked in a very cold location, for many hours, with no power applied.

(b) The temperature in the aircraft was low.

1) If the OFF light came on for this condition, let the temperature increase in the Equipment Cooling Bay to more than 32F (0C). Then do the Fault Isolation Procedure below.

B. Initial Evaluation

(1) Open and then close this circuit breaker: C1434, EQPT COOLING LOW FLOW DETECT EXHAUST [P18-3, A18].

(a) Wait a minimum of ten minutes while you monitor the equipment cooling exhaust OFF light.

(b) If the light went off in ten minutes or less, then there was an intermittent fault.

(c) If the light stays on after ten minutes, then continue.

(2) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

(3) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(4) If the light stays on after 20 seconds, then do these steps:

(a) Set the EQUIP COOLING EXHAUST switch to ALTN.

(b) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(c) Set the EQUIP COOLING EXHAUST switch to NORM.

EXPORT CONTROLLED:9E991
BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

(d) If the light went off in less than 20 seconds with the switch at ALTN, then do the Fault Isolation Procedure (Fan) below.

(e) If the light stayed on after 20 seconds with the switch at ALTN, then do the Fault Isolation Procedure (Airflow/Indication) below.

(5) If the light goes off in less than 20 seconds, then there was an intermittent fault.

C. Fault Isolation Procedure (Fan)

(1) Do this check of the circuits in the equipment cooling normal exhaust fan (WDM 21-27-31):

(a) Disconnect connector D160 from the equipment cooling normal exhaust fan, M98.

(b) For each of these pairs of pins of the connector on the equipment cooling normal exhaust fan, do a check for the specified result:

1) Pin 5 to pin 6, the circuit is good if there is continuity.

2) Pin 2 to pin 3, the circuit is good if there is continuity.

3) Pin 1 to pin 3, the circuit is good if there is continuity.

4) Pin 1 to pin 2, the circuit is good if there is continuity.

(c) If the circuit was not good for one or more of the checks, then do these steps:

1) Replace the equipment cooling normal exhaust fan, M98. These are the tasks:

- Equipment Cooling Exhaust Fan Removal, AMM TASK 21-27-04-000-801,
- Equipment Cooling Exhaust Fan Installation, AMM TASK 21-27-04-400-801.

2) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

3) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

4) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.

(d) If the circuit was good for all of the checks, then continue.

(e) Re-connect connector D160 to the equipment cooling normal exhaust fan, M98.

(2) Do this check of the continuity through the EQUIP COOLING EXHAUST switch (flight deck) (WDM 21-27-21):

(a) Disconnect the wires from the EQUIP COOLING EXHAUST switch (flight deck), S59.

(b) For each switch position below, set the switch to the specified position and do a check for continuity between the specified pins on the switch:

1) Switch position NORM, make sure there is continuity between wire terminal 2 and wire terminal 3.

(c) If there was a problem with the continuity, then do these steps:

1) Replace the EQUIP COOLING EXHAUST switch (flight deck), S59.

2) Re-connect the wires to the EQUIP COOLING EXHAUST switch (flight deck), S59.

3) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

4) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

5) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.

(d) If there was not a problem with the continuity, then continue.

(e) Re-connect the wires to the EQUIP COOLING EXHAUST switch (flight deck), S59.

(3) Do this check of the voltage at the equipment cooling normal exhaust fan:

(a) Disconnect connector D160 from the equipment cooling normal exhaust fan, M98.

(b) Do a check for 28 VDC between pins 5 and 7 of connector D160 (WDM 21-27-21).

(c) If 28 VDC is present, then do these steps:

1) Replace the equipment cooling normal exhaust fan, M98. These are the tasks:

- Equipment Cooling Exhaust Fan Removal, AMM TASK 21-27-04-000-801,
- Equipment Cooling Exhaust Fan Installation, AMM TASK 21-27-04-400-801.

2) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

3) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

4) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.

(d) If 28 VDC is not present, then continue.

BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

(4) Do this check of the wiring:
   (a) Do a wiring check to make sure that pin 7 of D160 has continuity to ground (WDM 21-27-21).
   (b) If you find a problem with the wiring, then do these steps:
      1) Repair the wiring.
      2) Re-connect connector D160 to the equipment cooling normal exhaust fan, M98.
      3) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.
      4) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
      5) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.
   (c) If you do not find a problem with the wiring, then continue.
   (d) Re-connect connector D160 to the equipment cooling normal exhaust fan, M98.

(5) Do this check of the voltage at the normal exhaust fan control relay:
   (a) Disconnect connector D1624 from the normal exhaust fan control relay, R29.
   (b) Do a check for voltage between pin X1 of connector D1624 and ground (WDM 21-27-21). The correct voltage is 28 VDC.
   (c) If the correct voltage is not present, then do this check of the voltage at the EQPT COOLING EXHAUST FAN CONT NORMAL circuit breaker:
      1) Get access to the EQPT COOLING EXHAUST FAN CONT NORMAL circuit breaker, C1435 [P6-4, C14].
      2) Do a check for voltage between terminal =L of the circuit breaker and ground (WDM 21-27-21). The correct voltage is 28 VDC.
      3) If the correct voltage is not present, then do these steps:
         a) Re-connect connector D1624 to the normal exhaust fan control relay, R29.
         b) Replace the EQPT COOLING EXHAUST FAN CONT NORMAL circuit breaker, C1435.
         c) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.
         d) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
         e) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.
      4) If the correct voltage is present, then do this check of the wiring:
         a) Do a wiring check between this pin of connector D1624 at the normal exhaust fan control relay and this terminal of circuit breaker C1435 (WDM 21-27-21):

D1624		**C1435**
pin X1	.....	term =L

         b) If you find a problem with the wiring, then do these steps:
            <1> Repair the wiring.
            <2> Re-connect connector D1624 to the normal exhaust fan control relay, R29.
            <3> Close the EQPT COOLING EXHAUST FAN CONT NORMAL circuit breaker, C1435 [P6-4, C14].
            <4> Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.
            <5> Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
            <6> If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.
   (d) If the correct voltage is present, then continue.

(6) Replace the normal exhaust fan control relay, R29.
   (a) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.
   (b) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
   (c) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.
   (d) If the light stayed on after 20 seconds, then continue.

(7) Do this check of the voltage supplied for the EQPT CLG EXH FAN PWR-NORM circuit breaker:
   (a) Get access to the EQPT CLG EXH FAN PWR-NORM circuit breaker, C836 [P91, E1].

BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

(b) Do a check for three-phase 115 VAC between terminals A2, B2, and C2 of the circuit breaker and ground (WDM 21-27-21).

(c) If three-phase 115 VAC is not present, then do these steps:

1) Replace the EQPT CLG EXH FAN PWR-NORM circuit breaker.

2) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

3) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

4) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.

(d) If three-phase 115 VAC is present, then continue.

(e) Close any access that you made to get to the EQPT CLG EXH FAN PWR-NORM circuit breaker, C836 [P91, E1].

(8) Do a check of the wiring between the EQPT CLG EXH FAN PWR-NORM circuit breaker, C836 and the equipment cooling normal exhaust fan, M98 (WDM 21-27-31).

(a) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN and then set it to NORM.

3) Wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

4) If the light is off, or if it went off in less than 20 seconds, then you corrected the fault.

D. Fault Isolation Procedure (Airflow/Indication)

(1) Open and then close this circuit breaker: C1434, EQPT COOLING LOW FLOW DETECT EXHAUST [P18-3, A18].

(a) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(b) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If the light stayed on after 20 seconds, then continue.

(2) Clean the equipment cooling airflow sensor. This is the task:

- Equipment Cooling Low Flow Sensors Cleaning, AMM TASK 21-27-03-400-802.

(a) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(b) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If the light stayed on after 20 seconds, then continue.

(3) Replace the equipment cooling airflow sensor, M100. These are the tasks:

- Equipment Cooling Low Flow Sensors Removal, AMM TASK 21-27-03-000-801,
- Equipment Cooling Low Flow Sensors Installation, AMM TASK 21-27-03-400-801.

(a) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(b) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If the light stayed on after 20 seconds, then continue.

(4) Do this check of the wiring:

(a) Disconnect connector D164 from the equipment cooling airflow sensor, M100.

(b) Open the EQPT COOLING LOW FLOW DETECT EXHAUST circuit breaker, C1434 [P18-3, A18].

BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

(c) Do a wiring check between this pin of connector D164 at the equipment cooling airflow sensor and this terminal of circuit breaker C1434 (WDM 21-27-21):

D164		C1434
pin 1	.....	term =L

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D164 to the equipment cooling airflow sensor, M100.
3) Close the EQPT COOLING LOW FLOW DETECT EXHAUST circuit breaker, C1434 [P18-3, A18].
4) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
5) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
6) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Close the EQPT COOLING LOW FLOW DETECT EXHAUST circuit breaker, C1434 [P18-3, A18].

(5) Do this check of the wiring:

(a) Disconnect connector D11418 from the power distribution panel 1, P91.

(b) Do a wiring check between this pin of connector D164 at the equipment cooling airflow sensor and this pin of connector D11418 at the power distribution panel 1 (WDM 21-27-21):

D164		D11418
pin 9	.....	pin 11

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D164 to the equipment cooling airflow sensor, M100.
3) Re-connect connector D11418 to the power distribution panel 1, P91.
4) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
5) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
6) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11418 to the power distribution panel 1, P91.

(6) Do this check of the wiring:

(a) Disconnect the wires from the equipment cooling exhaust OFF light, L47.

(b) Do a wiring check between this pin of connector D164 at the equipment cooling airflow sensor and these wire terminals at the equipment cooling exhaust OFF light (WDM 21-27-21):

D164		L47
pin 9	.....	term 2

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D164 to the equipment cooling airflow sensor, M100.
3) Re-connect the wires to the equipment cooling exhaust OFF light, L47.
4) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.
5) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

EXPORT CONTROLLED:9E991
BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

6) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D164 to the equipment cooling airflow sensor, M100.

(f) Re-connect the wires to the equipment cooling exhaust OFF light, L47.

(7) Replace the equipment cooling smoke auto reconfiguration timer relay, R1185.

(a) Set the EQUIP COOLING EXHAUST switch (flight deck) to ALTN, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(b) Set the EQUIP COOLING EXHAUST switch (flight deck) to NORM, then wait 20 seconds while you monitor the equipment cooling exhaust OFF light.

(c) If the light is off, or if it went off in less than 20 seconds with the switch at each position, then you corrected the fault.

--------- END OF CONTENT ---------

BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details